fective models and phenomenological theories of slow and glassy dynamics. The starting point are the ML methods, already described in Sec. III, that identify the local structure responsible for local relaxations on short time scales. The aim is now to use this structural field as a building block to construct a phenomenological model for how structural relaxation proceeds. This approach is built upon two useful classes of models to understand dynamics in glass-forming liquids and amorphous solids subjected to mechanical strain, namely trap and elasto-plastic models.

Trap models start from the high-temperature liquid, describing the system as a distribution of energy barriers for rearrangements. A recent implementation [105] adds a facilitation mechanism: in response to a rearrangement, all the energy barriers are subject to a small random drift. This facilitated trap model has been used to explain the emergence of excess wings in the relaxation spectra [106]. Elasto-plastic models start instead from the low-temperature solid, describing the system in terms of thermally-induced rearrangement events that can trigger other rearrangements via long-ranged strain fields [107–109]. Both of these approaches have given insights into glassy dynamics, and include facilitation effects in some manner, but both describe the local structure of the liquid in a very coarse-grained, simplistic manner. As a result, both classes of models must make ad hoc assumptions on the nature of the local relaxation events. In the case of the trap model, facilitation is assumed to lead to a shift of energy barriers, while in elasto-plastic models the distribution of yield stress is imposed.

A recent approach unites the trap and elasto-plastic models by extending them to include the local structure in the form of a machine-learned microscopic structural descriptor, the softness S, as introduced in Sec. III. Softness is sufficiently accurate that the probability of rearranging for particles of a given softness S, $P_R(S)$, has an Arrhenius temperature dependence: $P(R|S) = \exp\left[-\Delta E(S)/T + \Delta\Sigma(S)\right]$, where ΔE is the energy barrier and $\Delta\Sigma$ is the entropic one, $\Delta F = \Delta E - T\Delta\Sigma$. This suggests that particles of softness S have a well-defined free energy barrier for rearrangements, $\Delta F(S)$. The spatial variation of the softness then leads to a free-energy barrier field that couples to the stress and strain fields. Note that any local structural measure that predicts rearrangements or local yield stress [110] (whether extracted using ML, as described here or in other sections of this paper, or by other means [11]) could in principle be exploited in a similar manner to construct phenomenological models of glassy dynamics or plasticity.

Thanks to this ability to extract a free energy barrier estimate $\Delta F(S)$, the softness can naturally be used to construct a phenomenological trap model. However, softness allows one to go further by considering spatial correlations. When a rearrangement occurs, it alters the softness of rearranging particles as well as that of nearby particles through near-field facilitation [111]. It also alters the softness of more distant particles by creating a strain field that decays away from the rearrangement. Because the strain field changes the local structural environment of particles, it alters their softness. This far-field form of facilitation is well-captured by elasto-plastic models [108].

There is an interplay between rearrangements (strain), changes in softness and elasticity, with each one affecting the other two. A systematic approach to disentangling all of these effects has been introduced in Ref. [85] and implemented in a lattice structuro-elasto-plasticity model [31] for athermal systems under load. It has been applied successfully to a number of systems of varying ductility [31, 86] and used to extract insights into the microscopic factors that control ductility, such as the strength of near-field facilitation [86].

These results pave the way towards models of structural relaxation dynamics in glassy liquids. A simple trap model built upon the barriers $\Delta F(S)$ and assuming an underlying distribution of softness, $\rho(S)$, as in Ref. [112], was developed in Ref. [113]. One can also construct a version of the facilitated trap model of Ref. [105] that incorporates S. Generalising such models to supercooled liquids, however, is more challenging since one must include time-reversal invariance. Above the mode-coupling temperature, T_c, near-field facilitation should be sufficient. The hypotheses that $\rho(S)$ is nearly Gaussian [19] and that the near-field distribution of the change in softness $\Delta S(r)$ due to a rearrangement at the origin is also nearly Gaussian, are important simplifications that allows formulation of a closed theory [114]. For systems below T_c, however, it has been shown that long-ranged facilitation via strain occurs [111]. The inclusion of time-reversal invariance in a model such as a thermal elasto-plastic model [108] or structuro-elasto-plasticity model with long-ranged facilitation is a challenging open problem that needs to be solved. A precise predictor of future dynamics, as discussed in Sec. III, would be a very useful tool to adjust such models. Additionally, it would be interesting to explore whether the free energy barriers ΔF for local relaxation, extracted from the amorphous structure using softness, could be learned more directly and used to improve effective glasss models.

Another path forward is to switch from a field picture to a defect picture. Most predictors of rearrangements yield particle-based quantities that are easily converted to fields, but highlight localised regions that are susceptible to rearrangement [11]. These regions can be viewed as structural defects that interact with each other and are created and destroyed by strain and rearrangements. ML could be used to learn these interactions and rules, to help build defect theories of plasticity and glassy dynamics.

In this respect, a direction that has been recently explored consists in letting a glass-forming liquid evolve via the usual thermal motion (e.g., using molecular dynamics). During such exploration, the energy is periodically minimized in order to extract a library of mechanically stable zero-temperature configurations (inherent struc-

tures). The idea is then to use ML to classify pairs of inherent structures, instead of single configurations, and check whether the pair is connected by a low-energy excitation corresponding to a localised structural defect. For example, in Ref. [115], a large library of inherent structures was constructed by such an exploration at very low temperature inside a glass basin. By means of supervised learning techniques, it was possible to train a machine that inputs a pair of inherent structures and outputs, with good precision, the classical energy barrier separating them. This allowed for a speed up in the search for glass defects by more than one order of magnitude, which is significant given the complexity of these kind of calculations.

While this preliminary study was mostly focused on very low-energy defects that are associated with thermodynamic and transport anomalies of the glass at cryogenic temperatures [115], it should in principle be straightforward to extend the techniques to detect other kinds of defects, such as those associated with plastic events under shear [11, 116] of relaxation events under equilibrium thermal motion [117]. This is a promising direction for further research.

V. PERFORMANCE METRICS AND BENCHMARKING

An essential ingredient to fuel further development of ML techniques are detailed benchmarks for existing datasets, which may allow every researcher to independently develop and test new ML approaches without complex production and preprocessing of data [33]. We provide such benchmarks for ML glass-forming liquids for different systems, different dynamical observables, and different metrics.

The benchmarks are based on the dataset "GlassBench" that we have created and made publicly available (https://doi.org/10.5281/zenodo.10118191). The whole dataset is separated into a training set which, as the name suggests, can be used to train the neural network, and a test set which should be used only for benchmarking. In addition to initial amorphous structures and trajectories, we provide precalculated dynamical descriptors and propensities, as introduced in Sec. III. We have also uploaded sample python code for reading and processing. Additional technical information on the data format is provided with the dataset.

The tasks identified for "GlassBench" are directly related to the open questions highlighted in the introduction: (A) Train a model to predict single particle propensity purely from structural properties. Accuracy will be quantified using the Pearson correlation coefficient. Higher accuracy in the prediction indicates that the learned structural descriptor is indeed an important precursor for future relaxation but it can also become essential when using the model to generate new configurations. (B) Train a transferable model such that it can be accurately applied to different temperatures. This is an important task to enable investigation of structural relaxation at temperatures that are unreachable for numerical simulations. (C) Train a model which correctly predicts spatial dynamic heterogeneity (DH) as quantified by the dynamic susceptibility. The length scale of DH is growing with decreasing temperature such that at very low temperatures some regions in the system actively rearrange while others are completely frozen. DH are not only important for properties of glass-forming materials but are also core to fundamental theories of the glass transition [90, 118].

The ML techniques used for the benchmarking are summarized in Tab. I. In the following, we will refer to them just as models. A large variety of different models is represented, with very different numbers of fitting parameters and training time. Furthermore, the models use various ways to physically preprocess the structural input, either by using inherent states [30], or even by performing a Monte Carlo averaging of local cages [26]. These different factors, combined with the benchmarking provided below, should help choosing the most suitable method for a given purpose, with focus on either the highest-scoring predictions, computational efficiency, or interpretability. In addition to these ML techniques, we also include the performance of traditional structural descriptors based on physical intuition [9, 30, 119]. The model of glass-forming liquid that we selected to develop GlassBench is the very popular Kob-Andersen (KA) mixture, that we study both in two and in three dimensions.

Task A. The aim for the models is to learn correlations between the amorphous structure and the dynamic propensity of displacements, $\mathcal{R}_i(t) \equiv \langle \Delta \mathbf{r}_i(t) \rangle_{\rm iso}$, as introduced in Sec. III. A common metric used to assess the performance of different techniques is the Pearson correlation coefficient,

$$\rho_P = \frac{\mathrm{cov}(\mathcal{R}_i^{\rm MD}, \mathcal{X}_i^{\rm ML})}{\sqrt{\mathrm{var}(\mathcal{R}_i^{\rm MD})\mathrm{var}(\mathcal{X}_i^{\rm ML})}}, \tag{1}$$

between the labels $\mathcal{R}_i^{\rm MD}$ for each particle i of type 1 in the entire dataset as obtained from molecular dynamics (MD) simulations, and the ML output $\mathcal{X}_i^{\rm ML}$ [120]. The results are shown in Fig. 4a, with full symbols corresponding to supervised ML techniques (introduced in Sec. III) and open symbols to unsupervised techniques or physically-motivated structural descriptors (Sec. II). All technical details are provided in the SM [54]. From this figure, it can be concluded that supervised techniques nearly approach the maximal achievable correlation over the entire range of time scales. CAGE performs best for shorter times, which appears reasonable as it uses extensive Monte-Carlo simulations to characterize the local cage structure at short times. For longer times, the SE(3) GNN extension has the strongest correlation, closely trailed by the other advanced techniques. We can also observe that there is a pronounced gap between the supervised and the unsupervised techniques, indicating

	Training	ML approach	free parameters	states	training time	training hardware	app. time
BOTAN [25]	supervised	GNN	54200	th.	$\sim$ hours	NVIDIA A100	$\sim s$
CAGE [26]	supervised	Ridge regression	2775	th.+cage	$\sim$ min	CPU	$\sim$ hours
GlassMLP [30]	supervised	MLP	615	th.+inh.	$\sim$ min	CPU	$\sim s$
SE(3) [28]	supervised	GNN	52660	th.+inh.	$\sim$ hours	NVIDIA Tesla V100	$\sim s$
SBO [51]	unsupervised	PCA	0	th.	–	CPU	$\sim s$

TABLE I. Overview of the different techniques benchmarked in this roadmap. The column "states" refers to the usage of thermal (th.), inherent (inh.) or cage states. The training time corresponds to the time required to train one model at a specific temperature and time for the KA system in three dimensions. The application time corresponds to the time required to calculate the prediction for a single configuration, including preparation such as calculation of inh./cage states.

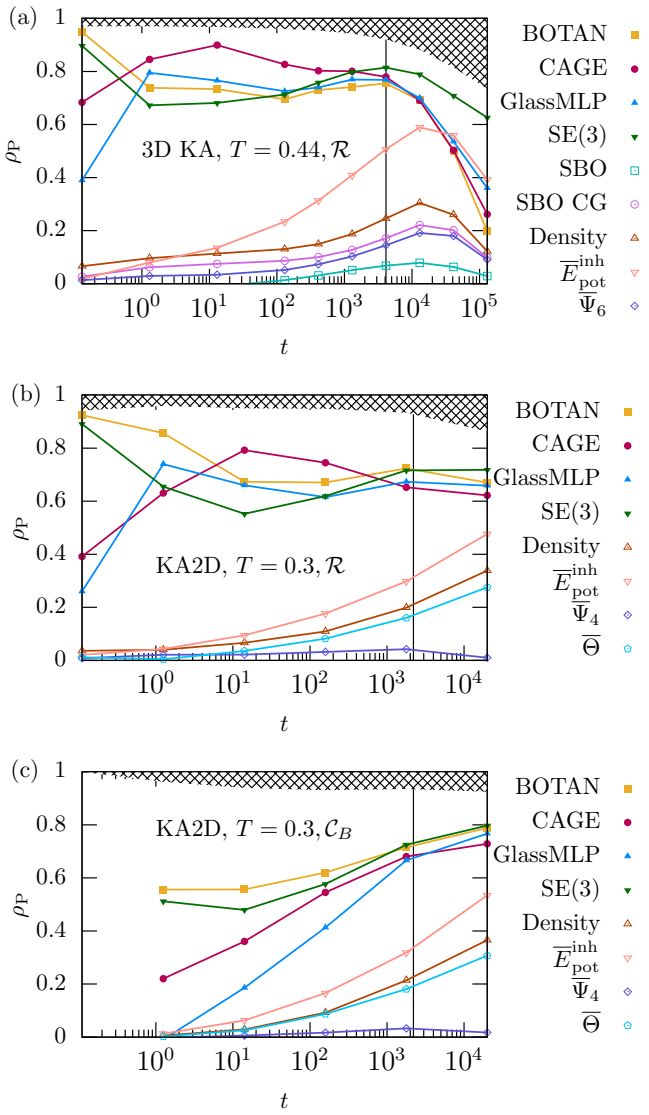

FIG. 4. **Benchmark task (A)** Pearson correlation ρ_P between various structural indicators with the ground truth. (a) 3D KA model at $T = 0.44$, (b,c) KA2D system at $T = 0.3$. The dynamical variable is (a,b) the propensity of displacements $\mathcal{R}$ and (c) the bond-breaking propensity $\mathcal{C}_B$. The vertical line marks the structural relaxation time τ_α, the typical time scale on which particles rearrange, as defined in Eq. (1) in the SM [54]. The exclusion zone on the top marks the highest achievable correlation given the finite number of replicas.

that the amorphous structural features that are predictive for dynamics do not stand out in a purely structural analysis, as discussed in Sec. II.

To asses the generality of these findings, we also provide benchmarks for a two-dimensional ternary mixture of Lennard-Jones particles (KA2D) in Fig. 4b. We find that, apart from minor differences, the performance of the individual techniques is very similar to the 3D system. The most noteworthy difference is perhaps that BOTAN performs best of all methods at the structural relaxation time $t \approx \tau_\alpha$ while in the 3D KA system, the trend is reversed. While there might be subtle differences between structural relaxation across spatial dimensions the above observation implies that the problem of learning correlations between structure and dynamics is essentially independent of the spatial dimension.

The propensity of displacements $\mathcal{R}_i$ is only one specific choice to characterise relaxation dynamics among many others. One alternative is to use the bond-breaking correlation, $\mathcal{C}_B^i$, which quantifies how many nearest neighbours particle i has lost during the relaxation process [30, 117]. As shown in Fig. 4c, the models also successfully learn correlations between the bond-breaking propensity and the amorphous structure. The performance of the models in the short-time predictions, however, is significantly reduced compared to the propensity of displacements. This implies that predicting the exact nature and position of the first rearrangement events appears more difficult than simply predicting short-time displacements. Around the structural relaxation time τ_α and beyond, the correlations shown in Fig. 4c for $\mathcal{C}_B^i$ are stronger than for $\mathcal{R}_i$ in Fig. 4b. This seemingly surprising result is connected to the growing dynamic heterogeneity at longer times (see also Fig. 6), which simplifies the prediction of larger rearranging clusters from coarse-grained structural properties, as discussed in Ref. [30]. Additionally, at times $t \geq \tau_\alpha$, we observe that the propensity of displacement $\mathcal{R}_i$ has slowly decaying tails which are not captured by the models (see Fig. S2 in the SM) and thus likely reduce the Pearson correlation.

Task B. An important property of supervised ML techniques is their transferability, in particular towards lower temperatures, with the goal to predict the dynamics at very low temperatures that are inaccessible by direct computer simulations [28, 89]. In Fig. 5, we evaluate the capabilities of the models to transfer their structure-